

Performance Analysis of LDPC Decoding: Hard Decision vs. Min-Sum Algorithms

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1 Introduction

The objective of this report is to analyze and compare the efficiency of two decoding methodologies for Low-Density Parity-Check (LDPC) codes: the Hard Decision (bit-flipping) decoder and the Soft Decision Min-Sum Algorithm (MSA). The analysis evaluates their performance over an Additive White Gaussian Noise (AWGN) channel using Binary Phase Shift Keying (BPSK) modulation.

2 System Model

The communication system is modeled using the following mathematical framework:

2.1 Modulation and Channel

The system utilizes BPSK modulation to map digital information bits $x \in \{0, 1\}$ to transmitted signals s :

$$s = \begin{cases} +1 & \text{if } x = 0 \\ -1 & \text{if } x = 1 \end{cases} \quad (1)$$

The signals are transmitted through an AWGN channel, where the received vector y is defined as $y = s + n$, and n is Gaussian noise with zero mean and variance σ^2 . In the initial test case, $\sigma^2 = 2$.

2.2 Parity-Check Matrix

The LDPC code is defined by the following 4×8 parity-check matrix H :

$$H = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \quad (2)$$

3 Decoding Methodologies

3.1 Hard Decision Decoding

The Hard Decision decoder makes bit estimates $\hat{x}_i$ based solely on the sign of the received signal:

$$\hat{x}_i = \begin{cases} 0 & \text{if } y_i > 0 \\ 1 & \text{if } y_i \leq 0 \end{cases} \quad (3)$$

The validity of the codeword is verified by computing the syndrome vector $S = H \cdot \hat{x}^T \pmod{2}$.

3.2 Min-Sum Algorithm (MSA)

The MSA is an iterative message-passing algorithm operating on a Tanner graph. It utilizes Log-Likelihood Ratios (LLR) initialized as $L(q_i) = \frac{2y_i}{\sigma^2}$.

Check Node Update ($M_{c \rightarrow v}$):

$$M_{r,c} = \left(\prod_{c' \in N(r) \setminus \{c\}} \text{sgn}(M_{c',r}) \right) \cdot \min_{c' \in N(r) \setminus \{c\}} |M_{c',r}| \quad (4)$$

Variable Node Update ($M_{v \rightarrow c}$):

$$M_{c,r} = L(q_i) + \sum_{r' \in M(c) \setminus \{r\}} M_{r',c} \quad (5)$$

The final decision is made based on the sign of the total LLR: $L_{total}(c) = L(q_i) + \sum_{r \in M(c)} M_{r,c}$.

4 Results and Analysis

4.1 Single Vector Case Study (Q.1)

The received vector $y = [0.2, 1.1, -0.3, 0.9, 0.7, -0.1, 1.2, 0.4]$ was processed through both decoders.

- **Hard Decision:** The thresholding produced $\hat{x} = [0, 0, 1, 0, 0, 1, 0, 0]$, which resulted in a syndrome of all zeros.
- **Min-Sum Algorithm:** The algorithm was forced to run for 3 iterations. Initial LLRs at indices 2 and 5 were -0.3 and -0.1 respectively. By the third iteration, these beliefs strengthened to -1.3 and -1.4 , reinforcing the error.

This result confirms that the noise at these indices perfectly matched the requirements of a valid (but incorrect) codeword, leading to a "successful" parity check despite remaining bit errors.

4.2 Monte Carlo Simulation (Q.2)

The simulation (1,000 trials per point) produced a classic performance curve. At 0 dB, the BER was observed at 0.03213. As the signal quality improved to 4 dB, the BER dropped significantly to 0.00013. Beyond 6 dB, the algorithm achieved a BER of 0.0, demonstrating the high reliability of the Min-Sum decoder under standard AWGN conditions.

5 Conclusion

This study successfully analyzed the performance trade-offs between Hard Decision and Min-Sum decoding methodologies for LDPC codes. The project highlighted two critical insights into iterative decoding theory:

- **Fundamental Limits of Coding:** The specific case study in Q.1 demonstrated that even robust algorithms like Min-Sum can be defeated by noise when the error pattern mimics a valid codeword. In this instance, the noise at indices 2 and 5 (considering zero-based indexing) was severe enough that the parity-check equations were satisfied, leading to a "successful" but incorrect decoding. This reinforces the concept that the minimum distance (d_{min}) of a code determines its ultimate error-correction limits.
- **The Power of Soft Information:** Conversely, the Monte Carlo simulation in Q.2 proved the clear statistical superiority of soft-decision decoding. While Hard Decision decoding suffers from a high error floor due to the loss of confidence information, the Min-Sum algorithm utilized Log-Likelihood Ratios (LLRs) to achieve a rapid "waterfall" drop in Bit Error Rate (BER). By the 6 dB threshold, the algorithm achieved a BER of 0.0, proving its resilience against standard random AWGN.

Ultimately, the iterative message-passing nature of the Min-Sum algorithm—where beliefs are refined over multiple iterations—remains a cornerstone of modern high-speed communications. While no algorithm can recover from a structural "trap" in the H matrix, the Min-Sum approach provides a critical balance between computational complexity and error-correction performance.